

Multi-Cloud Data Engineer Program (GCP + AZURE)

with Case Studies and Real-world Projects

Course duration: 4 months
Teaching Language: English
Mode: Online
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Cloud Introduction

- What is Cloud Computing?
- The need for cloud computing in modern businesses.
- Why organizations moved to Cloud
- Cloud Service Providers – GCP, Azure, AWS, ...

GCP Cloud Basics

GCP Cloud basics

- Introduction to Google Cloud Platform (GCP) and its core services
- GCP interfaces - Console, Cloud Shell and Cloud SDK
- GCP Locations - Regions, Zones & Global Infrastructure
- GCP Walkthrough - GCP Projects, Organizations, Folders & Billing Accounts
- GCP IAM & Admin - Principals, Roles, Service Accounts & Resource Hierarchy
- Setting up a Free Account in GCP

Linux Basics on Cloud Shell

- Getting started with Linux
- Linux Installation
- Basic Linux Commands
 - Cloud shell tips
 - File and Directory Operations (ls, cd, pwd, mkdir, rmdir, cp, mv, touch, rm,..)
 - File Content Manipulation (cat, less, head, tail, grep)
 - Text Processing (awk, sed, cut, sort, uniq)
 - User and Permission related (whoami, id, su, sudo, chmod, chown)

Python for Data Engineer

- Getting Started with Python
- Introduction to Python and basic functions
- Data Types – Strings, Operators, Numbers (Int, Float), Booleans
- Data Structures – Lists, Tuples, Dictionaries, Sets
- Python Programming Constructs - if, elif, else statements, for loops, while loops
- File I/O operations - Read, write, append, rename, remove, copy, move
- Exception Handling and Logging
 - what is exception handling, why it is needed
 - try and except blocks, else, finally, raise blocks
 - Why Logging? And Logging Levels
 - Writing Logs to File

- Data and Time operations - Working date and time, Formatting, Parsing, Time delta
- Modular Programming in Python
 - Functions & Lambda Functions
 - Classes
- REST API basics
 - What is API
 - HTTP methods (GET, POST, PUT, DELETE)
 - Status Codes, Request Headers, JSON Responses
 - Using the requests Library, Reading API Responses
- Modules and Packages
 - What is module and packages
 - Importing
 - Built-in modules
 - Install packages

GCP Data Engineering Tools

Google Cloud Storage

- Overview of Cloud Storage as a scalable and durable object storage service.
- Understanding buckets and objects in Cloud Storage.
- Use cases for Cloud Storage, such as data backup, multimedia storage, and website content
- Labs: using console & CLI to do below
 - Creating and managing Cloud Storage buckets.
 - Uploading and downloading objects to and from Cloud Storage.
 - Setting access controls and permissions for buckets and objects.
 - Data Transfer and Lifecycle Management
 - Versioning and Object Versioning
- Integration with Other GCP Services
- Monitoring and logging for Cloud Storage operations.

Cloud SQL

- Introduction to Cloud SQL
- Creating and Managing Cloud SQL Instances
- Configuring database settings, users, and access controls.
- Connecting to Cloud SQL instances using Cloud SQL studio, Shell, Workbenches
- Importing and exporting data in Cloud SQL.
- Backups and High Availability
- Integration with Other GCP Services
- Managing database user roles and permissions.
- Introduction to DMS
- End to End Database migration Project
 - Manual: Export and Import method
 - Automation: Cloud SQL DMS method

BigQuery (SQL development)

- Introduction to BigQuery
- BigQuery Architecture

- Use cases for BigQuery in business intelligence and analytics.
- Various method of creating table in BigQuery
- BigQuery Data Sources and File Formats
- Native table and External Tables
- Working with complex data types
 - Working json data, nested, repeated and array data
- Data Integration and Export
 - Loading data into BigQuery from Cloud Storage, Cloud SQL, and other sources.
 - Exporting data from BigQuery to various formats.
 - Real-time data streaming into BigQuery.
- Configuring access controls and permissions in BigQuery.
- BigQuery Views:
 - Views
 - Materialized Views
 - Authorized Views
- Optimization techniques in BigQuery
- BigQuery Slots – on demand, flat-rate, flex-slots
- *Case Study-1: implement a real-world analytics data platform for Spotify*
- *Case Study-2: Enterprise Social Media analytics platform*

DataProc (Pyspark Development)

- Introduction to Hadoop and Apache Spark
- Understanding the difference between Spark and MapReduce
- What is Spark and Pyspark.
- Understanding Spark framework and its functionalities
- Overview of DataProc as a fully managed Apache Spark and Hadoop service.
- Cluster Creation and Configuration
 - Creating and managing DataProc clusters.
 - Configuring cluster properties for performance and scalability.
 - Preemptible instances and cost optimization.
- Learning Pyspark:
 - How to read from multiple data sources
 - How to perform multiple transformations
 - How to write to multiple targets
- Running Jobs on DataProc
 - Submitting and monitoring Spark and Hadoop jobs on DataProc.
 - Use of initialization actions and custom scripts.
 - Job debugging and troubleshooting.
- *Case study-1: Data Cleaning of Employee Travel Records*
- *Case study-2: Processing real-time patient health data*
- *Case study-3: Creating a pyspark job to support ML model creations*

DataFlow (Apache Beam development)

- Introduction to DataFlow
- Use cases for DataFlow in real-time analytics and ETL.
- Understanding the difference between Apache Spark and Apache Beam
- How Dataflow is different from Dataproc
- Learning Apache Beam

- How to read from multiple data sources
- How to perform multiple transformations
- How to write to multiple targets
- *Case study-1: Template method of creating pipelines*
- *Case study-2: E-commerce Transaction Processing*
- *Case study-3: End to End Streaming Pipeline using Apache beam with Dataflow, Python app, PubSub, BigQuery, GCS*

Cloud Pub/Sub (Streaming)

- Introduction to Pub/Sub
- Understanding the role of Pub/Sub in event-driven architectures.
- Key Pub/Sub concepts: topics, subscriptions, messages, and acknowledgments.
- Creating and Managing Topics and Subscriptions
 - Using the GCP Console to create Pub/Sub topics and subscriptions.
 - Configuring message retention policies and acknowledgment settings.
- Publishing and Consuming Messages
 - Writing and deploying code to publish messages to a topic.
 - Implementing subscribers to consume and process messages from subscriptions.
- *Case study-1: Streaming use-case using Dataflow*

Cloud Composer (DAG Development)

- Introduction to Composer/Airflow
- Overview of Airflow Architecture
- Use cases for Composer in managing and scheduling workflows.
- Creating and Managing Workflows
 - Creating and configuring Composer environments.
 - Defining and scheduling workflows using Apache Airflow.
 - Monitoring and managing workflow executions.
- Integration with Data Engineering Services
 - Orchestrating workflows involving BigQuery, DataFlow, and other services.
 - Coordinating ETL processes with Composer.
 - Integrating with external systems and APIs.
- Error Handling and Troubleshooting
 - Handling errors and retries in Composer workflows.
 - Debugging and troubleshooting failed workflow executions.
 - Logging and monitoring for Composer workflows.
- *Level-1-DAG: Orchestrating the BigQuery pipelines*
- *Level-2-DAG: Orchestrating the DataProc pipelines*
- *Level-3-DAG: Orchestrating the Dataflow pipelines*
- *Deploy DAGs: Implementing CI/CD in Composer Using Cloud Build and GitHub*

Azure Data Engineering Tools

Azure Introduction

- Introduction to Azure Cloud Platform and its core services
- Azure interfaces - Console, Cloud Shell and Cloud SDK

- Azure Locations - Regions, Zones & Global Infrastructure
- Azure Walkthrough – Subscriptions, Resources, Resource group & Management group
- Azure IAM & Admin - Identities, Roles, Service Accounts & Resource Hierarchy
- Setting up a Free Account in Azure Cloud

Azure DataLake Storage (ADLS)

- Introduction to Azure Data Lake Storage (ADLS)
- Core Components of ADLS Gen2
 - Blobs, Containers, and Storage Accounts
 - Hierarchical Namespace
- Why Use Azure Data Lake Storage?
 - Scalability, Cost-Effectiveness, Security
 - Integration with Analytics Services
- Access and Management of ADLS
 - Data Management via Console, SDK, CLI, REST API
 - Role-Based Access Control (RBAC) and Access Control Lists (ACLs)
- Pricing Overview
 - Pricing Tiers (Hot, Cool, Archive)
 - Cost Factors
- Best Practices for ADLS
 - Data Partitioning
 - Tiering Strategy for Cost Optimization
 - Monitoring and Usage Tracking
- Interactive Access Methods
 - Managing ADLS via Console
 - CLI/Shell Interaction
 - Programmatic Access via SDKs

Azure SQL DB

- Introduction to Azure SQL Database
- Key Features
- Creating a SQL db for practice

Azure Data Factory (ADF)

- What is ADF and Use cases
- Core components of ADF and understanding & configuring
 - Integration runtime
 - Linked Services
 - Datasets
 - Pipelines
 - Activities
 - Triggers
- Key Features of Azure Data Factory
- Pipelines in ADF
 - Creating and Configuring Pipelines
 - Activities and Triggers
 - Monitoring and Managing Pipeline Runs
 - Orchestration

- Multiple examples with ADF activities
- *Case study-1: End to End Incremental pipeline with ADF, ADLS, SQL DB.*
- Best Practices for ADF
 - Designing Efficient Pipelines
 - Data Partitioning and Parallelism
- Monitoring and Optimization for Performance

Azure Synapse

- Synapse Workspace, SQL Pools & Architecture
- Dedicated & Serverless SQL concepts
- Data ingestion using Synapse Pipelines
- External Tables, Views & Data Lake integration
- Query performance optimization, partitioning & workload management

Databricks on Azure/GCP

- Introduction to Databricks Lakehouse Platform
- Databricks Architecture and Core Components
 - Explore the architecture, including clusters, jobs, and the workspace environment.
 - Key components like Databricks Runtime, Spark, and Delta Lake.
- Setting Up and Administering a Databricks Workspace
 - Step-by-step guide on workspace creation and management.
 - Admin tasks including user management, permissions, and billing.
- Working with Notebooks and Clusters
 - Collaborative data science using Databricks notebooks.
 - Cluster setup, management, and autoscaling.
- Databricks Unity Catalog
 - Unified governance solution for managing data and metadata across the lakehouse.
 - Organizing data assets and enforcing security policies.
- Managing Data with Delta Lake
 - Understanding Delta Lake's ACID transactions and versioning capabilities.
 - Best practices for efficient data management.
 - Creating Delta Lake Tables
 - Querying and managing data across multiple file formats.
- Building ELT Pipelines with Spark SQL and Python
- Incremental Data Processing with Delta Lake
- Case Study 1: Real-time data processing with Spark Structured streaming
- Case Study 2: Databricks Autoloader for file-based ingestion utility
- Performance Optimization in Databricks

By the End of the course What Students can Expect

1. Build end-to-end Data Engineering pipelines on GCP & Azure
2. Develop strong skills in SQL, Python, PySpark, Linux & Apache Beam
3. Work with industry standard data engineering services
4. Implement Batch & Real-Time Data Processing pipelines
5. Design Bronze → Silver → Gold / Medallion Architecture
6. Build production-ready projects and troubleshoot real-world data engineering scenarios

7. Understand Cloud, Data Lake, Data Warehouse & Lakehouse architectures
8. Prepare for Data Engineer interviews with practical, scenario-based knowledge

Thank You.
Shaik Saidhul
Cloud Expert